

Biomass emitter plug-in – Quick Reference Guide

Version 0.1, 30 May 2018, JS

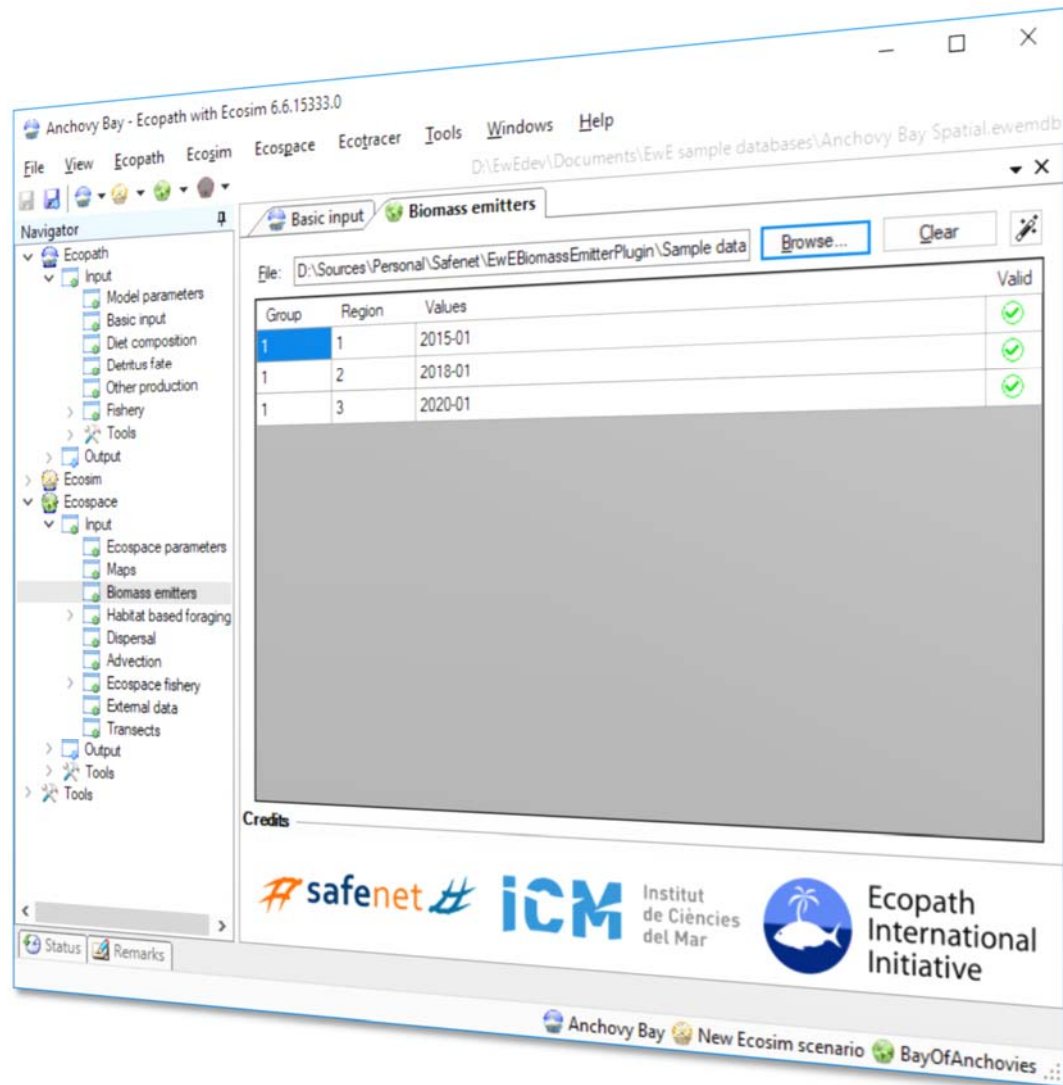


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1 How does it work

Dedicated models of MPA can only be used to adequately represent the local effects of MPAs, as these models are created at sufficient spatial scales to allow species to thrive and move within the MPA while not getting caught. Coarser models do not have the cell resolution to reflect these dynamics. This has been a limiting factor to deciding on spatial scales of EwE models.

The biomass emitter is a light-weight plug-in to the Ecopath with Ecosim food web modelling desktop software, intended to transfer biomass effects of local features, obtained from dedicated fine-scale spatial models, to models of coarser spatial scales that otherwise cannot reproduce the impact of these local features. An obvious application of the biomass emitter is to transfer the local benefits of a MPA to a regional model (Figure 1).

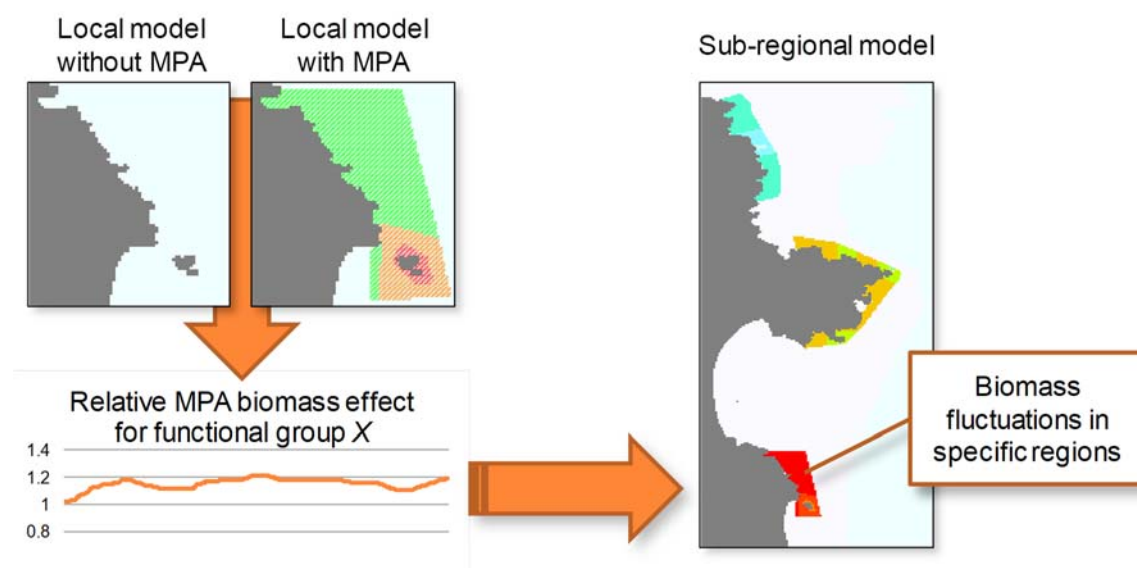


Figure 1 - The Biomass Emitter plug-in transfers relative biomass trends from regions in fine-scaled models to those same regions in coarse-scaled models. This enables regional models to integrate local predictions

The emitter transfers this information using time series of relative biomass effects for selected functional groups, bridging the data across models through pre-assigned regions. If both models define the same regions, but at different spatial resolutions, relative biomass fluctuations of specific groups in a fine model can be brought to a coarser model.

The Biomass Emitters plug-in reads CSV files of the following format:

group	1	1	1
region	1	2	3
2015-01	2		
2018-01		2	
2020-01			2

Each row identifies relative biomass fluctuations for a single group and region, over time. Time is defined in absolute terms as 'year-month', and emission values are allowed to be missing.

When Ecospace runs, for every time that a biomass emission is defined, biomasses in all cells that belong to an indicated region will be multiplied by the emission amount. It's that simple.

2 Generating Biomass Emission CSV input files

The CSV file needed by the Biomass Emitters plug-in can be created by hand for exploring its impact. The plug-in is intended to incorporate predictions of fine spatial resolution into models of coarser resolution; a utility is included in the plug-in that will create this file from regional average files that Ecospace can save.

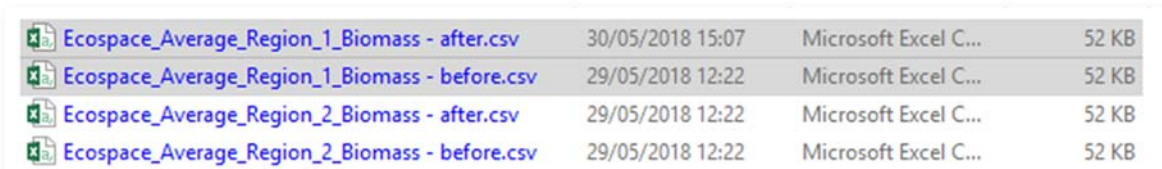
To generate a Biomass Emission input CSV file with the included utility, make sure you:

- Have regions defined for each MPA. If you use the Define Regions > Region for each MPA tool, a region will be defined for each MPA, numbered to the sequential index of an MPA. Note that for best results MPAs should not overlap, as regions cannot overlap.
- Have Ecospace set to auto-save region averages (see Ecospace > input > parameters)

The idea is now to run Ecospace twice, once with relevant MPAs enabled, and once with these MPAs disabled. The utility will use the difference between the two auto-saved region averages to create the biomass emission CSV file for the coarser resolution model.

- Run Ecospace model with your first MPA scenario configured (e.g., no MPAs enforced). After the model run, rename the relevant region average CSV file that their name ends with “-before.csv”
- Run Ecospace model again with your second MPA scenario configured (e.g. all MPAs enforced). After the model run, rename the relevant region average CSV file that their name ends with “-after.csv”

The utility looks for the ‘before’ and ‘after’ classifiers to determine which file comes first, and which second, when calculating the Biomass Emissions (see Figure 2).



Ecospace_Average_Region_1_Biomass - after.csv	30/05/2018 15:07	Microsoft Excel C...	52 KB
Ecospace_Average_Region_1_Biomass - before.csv	29/05/2018 12:22	Microsoft Excel C...	52 KB
Ecospace_Average_Region_2_Biomass - after.csv	29/05/2018 12:22	Microsoft Excel C...	52 KB
Ecospace_Average_Region_2_Biomass - before.csv	29/05/2018 12:22	Microsoft Excel C...	52 KB

Figure 2 - Example renamed region average files, ready to be passed to the Biomass Emission utility

The process to generate the Biomass Emission CSV file is simple:

- Launch the utility from the Biomass Emission plug-in by clicking the magic button (✂). This brings up the utility as shown in Figure 3;
- Drag and drop any number of ‘before’ and ‘after’ file pairs onto the designated area in the utility;
- Select for which region / group pairs you would like to generate the emission file;
- Enter or browse to the emission output file location;
- Save.

If successful, the newly generated emission file will be loaded in the Biomass Emission user interface.

Biomass emitter CSV generation utility

Drop region average files here
(both -before.csv and -after.csv)

	Group	Region 1	Region 2
1	Whales	☒	☐
2	Seals	☐	☒
3	Cod	☐	☒
4	Whiting	☐	☐
5	Mackerel	☐	☐
6	Anchovy	☐	☐
7	Shrimp	☐	☐
8	Benthos	☐	☐
9	Zooplankton	☐	☐
10	Phytoplankton	☐	☐
11	Detritus	☐	☐

Save to:

Figure 3 - The Biomass Emission CSV input file creation utility